

Consider the wave eq

$$u_{tt} = 9 u_{xx} \Rightarrow c = 3$$

$$-2 \leq x \leq 2, t \geq 0, \quad (1a)$$

$$u(-2, t) = u(2, t) = 0$$

$$t \geq 0 \quad (1b)$$

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x)$$

$$-2 \leq x \leq 2. \quad (1c)$$

a) Find the exact solutions of the wave eq for the following two sets of initial cond.

$$f(x) = \sin\left(\frac{\pi x}{2}\right) \quad \text{and} \quad g(x) = 0. \quad (2)$$

$$f(x) = 0 \quad \text{and} \quad g(x) = \pi \cos\left(\frac{\pi x}{2}\right) \quad (3)$$

$$(2): u(x, t) = F(x) G(t)$$

$$F'' - kF = 0 \Rightarrow F'' + p^2 F = 0, \quad k = \frac{u}{u} = -p^2$$

$$G'' - kG = 0$$

$$F(x) = A \cos(px) + B \sin(px)$$

$$F(-2) = A \cos\left(-2\frac{\pi}{2}\right) + B \sin\left(-2\frac{\pi}{2}\right) = 0$$

$$F(-2) = A \cos\left(-\frac{\pi}{2}\right) + B \sin\left(-\frac{\pi}{2}\right) = 0 \quad \text{IDK}$$

$$F(2) = B \sin(4p) = 0, \quad B \neq 0 \Rightarrow B = 1$$

$$4p = \pi n \Rightarrow p = \frac{\pi n}{4} \Rightarrow k = -p^2 = -\left(\frac{\pi n}{4}\right)^2$$

$$G'' + \left(\frac{\pi n}{4}\right)^2 G = 0$$

$$G(t) = C_n \cos(pt) + D_n \sin(pt)$$

$$u_n(x, t) = \sin\left(\frac{n\pi x}{4}\right) [C_n \cos(pt) + D_n \sin(pt)]$$

$$u(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{4}\right) [C_n \cos(pt) + D_n \sin(pt)]$$

$$u(x, 0) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{4}\right) = \sin\left(\frac{\pi x}{2}\right) \Rightarrow n=2, C_2=1, C_n=0 \text{ hvis } n \neq 2$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} D_n \lambda_n \sin\left(\frac{n\pi x}{4}\right) = 0$$

$$u(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{4}\right) [C_n \cos\left(\frac{\pi n}{4} t\right) + D_n \sin\left(\frac{\pi n}{4} t\right)]$$

$$= \sin\left(\frac{n\pi x}{4}\right) C_n \cos\left(\frac{\pi n}{4} t\right) + \sin\left(\frac{n\pi x}{4}\right) D_n \sin\left(\frac{\pi n}{4} t\right)$$

$$= \sin\left(\frac{\pi x}{2}\right) \cos\left(\frac{\pi t}{2}\right)$$

$$C_2=1, C_n=0 \text{ hvis } n \neq 2$$

Prob. 2 - D'Alembert's

Ex 8 - K.P. Lind

Use D'Alembert's method to find the solution of the wave eq.

$$\partial_t^2 u = 4 \partial_x^2 u \Rightarrow c = 2$$

for $t > 0$ and all $x \in \mathbb{R}$ with initial conditions

$$u(x, 0) = \sin(x) \quad \text{and} \quad u_t(x, 0) = e^{-\frac{x}{4}}.$$

The solution to the wave eq. with D'Alembert's method is given by

$$u(x, t) = \frac{1}{2} (f(x+ct) + f(x-ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\tilde{x}) d\tilde{x}$$

$$u(x, 0) = \sin(x) = f(x) \quad u_t(x, 0) = e^{-\frac{x}{4}} = g(x)$$

$$= \frac{1}{2} (f(x+2t) + f(x-2t)) + \frac{1}{2 \cdot 2} \int_{x-2t}^{x+2t} g(\tilde{x}) d\tilde{x}$$

$$= \frac{1}{2} (\sin(x+2t) + \sin(x-2t)) + \frac{1}{4} \int_{x-2t}^{x+2t} e^{-\frac{\tilde{x}}{4}} d\tilde{x}$$

$$= \frac{1}{2} (\sin(x+2t) + \sin(x-2t)) - \frac{1}{4 \cdot 4} [e^{-\frac{\tilde{x}}{4}}]_{x-2t}^{x+2t}$$

$$= \frac{1}{2} (\sin(x+2t) + \sin(x-2t)) - \frac{1}{16} (e^{-\frac{x}{4} + \frac{t}{2}} - e^{-\frac{x}{4} - \frac{t}{2}}) \quad \sin x \cos y = \frac{1}{2} [\sin(x+y) + \sin(x-y)]$$

$$= \underline{\sin(x) \cos(2t) + \frac{1}{16} (e^{\frac{t}{2} - \frac{x}{4}} - e^{-\frac{t}{2} - \frac{x}{4}})}$$

Prob. 3 - 1D Heat eq

Solve the differential eq

$$\begin{cases} \partial_t u - \frac{1}{4} \partial_x^2 u = 0, & \Rightarrow c = \frac{1}{2} \\ u(0, t) = u(\frac{\pi}{2}, t) = 0, \\ u(x, 0) = 2x(x - \frac{\pi}{2}) \end{cases}$$

for $x \in [0, \frac{\pi}{2}]$ and $t \in [0, \infty]$

$$u(x, t) = F(x)G(t)$$

$$F'' - kF = 0 \Rightarrow F'' + p^2 F = 0$$

$$G'' - kG = 0$$

$$F(x) = A \cos(px) + B \sin(px)$$

$$F(0) = A = 0 \Rightarrow A = 0$$

$$F(\frac{\pi}{2}) = B \sin(\frac{\pi}{2}p) = 0, \quad B \neq 0 \Rightarrow B = 1$$

$$\frac{\pi p}{2} = \pi n \Rightarrow p = 2n \Rightarrow k = -p^2 = -4n^2$$

$$G'' + 4n^2 G = 0$$

$$G(t) = B_n e^{-4n^2 t}$$

$$u_n(x, t) = B_n e^{-4n^2 t} \sin(2nx)$$

$$u(x, t) = \sum_{n=0}^{\infty} B_n e^{-4n^2 t} \sin(2nx)$$

$$u(x, t) = \sum_{n=0}^{\infty} B_n \sin(2nx) = f(x) = 2x(x - \frac{\pi}{2})$$

$$B_0 = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} f(x) dx = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} 2x(x - \frac{\pi}{2}) dx = \frac{1}{\pi} \left[\frac{2x^2}{2} - \frac{\pi x^2}{2} \right]_0^{\frac{\pi}{2}} = \frac{1}{\pi} \left(\frac{\pi^2}{8} - \frac{\pi^3}{8} \right) = \frac{\pi^2}{8} - \frac{\pi^3}{8} = \frac{\pi^2}{6}$$

$$B_n = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} f(x) \sin(2nx) dx \dots$$

$$u(t, x) = \frac{\pi^2}{6} + \sum B_n \sin(2nx)$$